Care Flow: A Bilingual Digital Queue Management System for Outpatient Services at AIC Kijabe Hospital

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Introduction

The outpatient department at AIC Kijabe Hospital managed daily patient flow through manual, paper-based queues. Overcrowded waiting areas produced long and unpredictable waits, queue-jumping disputes, and missed calls, and staff had no auditable record of the patient journey from arrival to consultation.

English-only signage and announcements excluded Kiswahili-speaking patients, while cash-only payment steps fragmented the visit record: registration, payment and clinical care were documented in separate, disconnected paper trails.

Patient waiting time is a documented determinant of satisfaction with outpatient care (Oche & Adamu, 2013). The World Health Organization recommends digital health interventions for health-system strengthening (WHO, 2019), and Kenya's National eHealth Policy 2016–2030 calls for people-centred digital health services. These priorities motivate a locally designed, bilingual digital queue for outpatient care.

OBJECTIVE

To design, build and deploy a secure, bilingual electronic queue system covering the full outpatient journey — registration → payment → triage → consultation — with live room displays and voice announcements.

Methods

- **Setting & design.** AIC Kijabe Hospital outpatient department; iterative design–build–deploy cycles during 2026; registration-first patient flow.
- **Queue engine.** PostgreSQL security-definer daily sequences issue atomic token numbers — duplicate tokens are impossible by construction; tokens reset daily.
- **Payments.** M-Pesa Daraja STK push to the patient's phone, with insurance-provider verification as an alternative path; payment confirmed before triage.
- **Patient flow.** Bilingual self-check-in kiosk (English / Kiswahili; new-patient check-in is Kiswahili-first) → triage → waiting → room call.
- **Calling.** Live display board covering **57 active consultation rooms**; public-address announcements generated with Gemini text-to-speech (**3 voices**, double chime + token read twice, with browser-speech fallback).
- **Staff tools.** Role-based dashboards, biometric work-ID sign-in (WebAuthn), room assignments, attendance and summons.
- **Security.** 12 layered controls including Cloudflare Turnstile, database-backed rate limiting (5 attempts / 10 min → 15 min block), row-level security, and an append-only audit trail.
- **Data lifecycle.** Nightly 00:00 (Africa/Nairobi) archive of completed visits to Excel (.xlsx) workbooks auto-delivered to the superadmin's device; patients still waiting or in rooms are never purged.
- **Stack.** React 19 + TanStack Start + TypeScript; PostgreSQL on Neon; hosted on Render; installable offline-capable PWA.

Results

The complete system is deployed and running end to end in production. Queue integrity is enforced at the database level; every completed visit is archived nightly and auto-exported to Excel. Formal waiting-time and satisfaction outcomes are the planned next phase of work.

Verified system scope

57 active rooms	**22** services	**23** relational tables
12 security controls	**2** interface languages	**3** PA voices

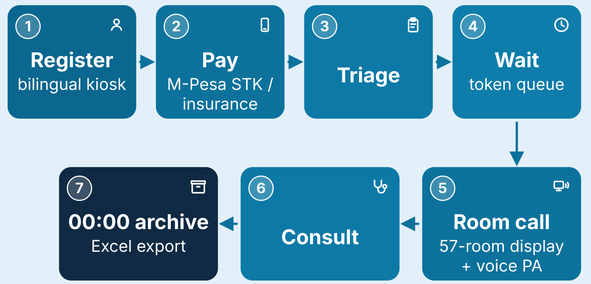


Figure 1. Patient journey through the Care Flow system, from bilingual registration to nightly Excel archive.

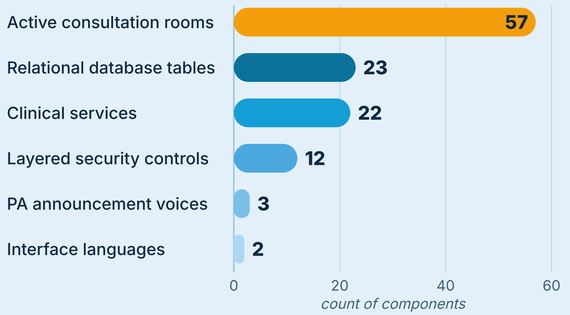


Figure 2. Verified scope of the deployed system (component counts, November 2026).

Conclusions

A low-cost, locally built digital queue system is feasible for a Kenyan mission hospital using standard web technologies. Atomic tokens make the queue fair by construction; Kiswahili-first check-in makes it language-inclusive; M-Pesa integration keeps payment inside the visit record; and nightly Excel archives make the data auditable.

Limitations. Development took place at a single site, and waiting-time and patient-satisfaction outcomes have not yet been measured.

Next steps. The next phase is a prospective before–after waiting-time study with a patient-satisfaction survey, followed by scale-up to partner facilities.

Acknowledgements. AIC Kijabe Hospital staff and patients, and the KCHS Research Committee.

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